

Dr. Saurabh Joshi

Director of Engineering & Applied Researcher

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 LinkedIn
 GitHub / Company

PROFESSIONAL EXPERIENCE

Consulting Advisor *Cerify.ai*

Jul 2026–Present Remote

- **Strategic Advisory:** Advising leadership on technological roadmap in the security and audit space, shaping product direction for automated vulnerability detection.
- **Engineering Mentorship:** Mentoring the entire engineering team on architecting solutions for automated vulnerability detection engines, bringing formal methods and static analysis expertise to core detection capabilities.

Director of Engineering and Innovations *Supra*

Oct 2024–Present Remote

- **Organizational Leadership:** Lead 4 product engineering teams (Smart Contracts, Execution Layer, Oracle, Automation — 14+ engineers) and coordinate cross-functionally with Layer 1, Wallet, DeFi, and Explorer teams to deliver integrated platform releases.
- **Product Conceptualization (0 → 1):** Conceived and drove the creation of a native event-driven automation platform, eliminating dependency on external off-chain services. Leveraged existing VM execution paths to accelerate delivery, achieving zero-block-delay execution guarantees and scaling to 50,000 concurrent automation tasks.
- **High-Performance Data Platform:** Steered the architecture of the core real-time data distribution protocol (DORA), delivering sub-second latency for 600+ data feeds per second, adopted across 80+ partner networks.
- **Quality & Engineering Culture:** Instituted code coverage enforcement, PR review checklists and templates, and championed a culture of unit testing, property-based testing, and modular, reusable design across all teams. Pioneered AI-assisted test generation to maintain velocity under resource constraints.
- **Infrastructure Optimization:** Recommended PostgreSQL schema modifications that reduced database storage requirements by 20%, directly lowering operational costs.
- **Security & Incident Response:** Led incident response for multiple critical security events — analyzing blast radius, coordinating cross-team fixes, and executing expedited production releases. Established 24/7 operational coverage through DevOps on-call rotations, emergency escalation chains, and automated liveness monitoring with alerting.
- **Governance & Partner Relations:** Served as a trusted member of the multi-signature governing authority managing protocol-level security and upgrades. Engaged directly with partners and enterprise clients to define integration requirements and drive adoption.
- **Executive Communication:** Reported directly to the CEO on technical strategy, requirements, and crisis response. Represented the company at developer events and authored technical papers and blog posts.

Principal Engineer *Supra*

May 2022–Sep 2024 Remote

- **Engineering Rescue & Turnaround:** Assumed technical ownership of the DORA oracle protocol after the departure of the previous engineering director. Inherited a partial, incorrect, and non-modular implementation; redesigned the architecture for correctness, modularity, and cross-network compatibility.
- **Novel Systems Design:** Invented a merkle-tree chunking scheme enabling a single cryptographic signature over 600+ data feeds to be distributed to networks with heterogeneous payload size constraints — reducing bandwidth and verification costs. Optimized provenance data flow by offloading responsibility to data-gatherer nodes, cutting storage and network overhead.
- **Consistent Cross-Platform APIs:** Designed and shipped a unified API and client SDK layer consumed across multiple runtime environments, ensuring a consistent developer experience for external integrators.
- **Research to Production:** Pioneered the mathematical design of the DORA consensus protocol, securing a US Patent and publishing a peer-reviewed paper at IEEE ICDCS 2023.
- **Formal Verification Strategy:** Initiated the company's formal verification framework and contract specification strategy to mathematically prove correctness of critical execution paths.

Staff Engineer, Product Security *Qualcomm*

Oct 2021–May 2022 Hyderabad, India

- **Team Leadership (14 Engineers):** Led and managed a Security Validation team of 14 engineers designing proof-of-concept exploits and security mitigations for the mobile software stack. Validation work directly secured chipsets shipped in commercial mobile devices.
- **Process Optimization (70% Faster PR Cycles):** Diagnosed a systemic code review bottleneck caused by context-switching and deprioritized reviews. Engineered an automated metrics pipeline using the GitHub API to surface developer and reviewer dwell times, sending transparent reports to the team — reducing PR cycle time by 70%.
- **Talent Pipeline:** Cultivated strategic university relations across India to build a high-caliber talent pipeline. Personally involved in hiring, interviewing, and recommending engineers who became strong contributors.
- **People Management:** Directed performance evaluations, career development, and mentorship programs for the combined engineering team.

Assistant Professor Grade I *Department of CSE, IIT Hyderabad*

May 2016–Sep 2021 Hyderabad, India

- **Applied R&D Leadership:** Headed a research group (2 PhD students, 4 Masters students, and undergraduate researchers) pursuing applied systems research in program analysis, compiler verification, software security, and constraint optimization.
- **Grant Acquisition & Budget Management:** Secured grants totaling over ₹4.9 Crore (\$600K) in competitive research funding and industry consultancies from national agencies (DST, NSCS, Ministry of Defense, CDAC) and international corporations (Suzuki Motor Corporation, Japan).
- **Industry Consultancies:**
 - *Suzuki Motor Corporation, Japan:* Consulted on optimizing build times and binary sizes for Autograde Linux systems.
 - *Ministry of Defense:* Developed MISRA C code scanners for safety-critical embedded software compliance.
 - *NSCS:* Contributed to formulating security standards for operating systems and firmware in national network infrastructure devices.
- **Award-Winning Open-Source Tools:** Developed and open-sourced Pinaka (symbolic execution engine), Open-WBO-Inc (optimization solver), and GPURepair (automated repair of concurrency bugs in GPU kernels). Tools won multiple Gold/Silver medals at international solver competitions.
- **Teaching:** Delivered 45 credits of graduate and undergraduate courses including Software Verification, Constraint Programming, Principles of Programming Languages, and Introduction to Blockchain Programming.

Assistant Professor (Grade I) *Department of CSE, IIT Guwahati*

Sep 2015–May 2016 Guwahati, India

- Conducted research and directed graduate thesis projects in computer science and systems engineering.

Post Doctoral Research Fellow *Department of Computer Science, University of Oxford* **Aug 2012–Aug 2015** Oxford, UK

- Developed formal verification frameworks (2LS, cbmc-repair, Open-WBO) under major international research initiatives, collaborating with academic and industrial partners across Europe.

Research Intern *Rigorous Software Engineering group, Microsoft Research India*

May 2010–Apr 2011 Bangalore, India

- Developed automated static analysis tools for detecting concurrency bugs and under-specified software harnesses in enterprise systems. Work resulted in a US Patent and a publication at POPL 2012.

Project Trainee *Software Engineering group, IBM India Research Lab*

Aug 2006–Jul 2007 New Delhi, India

- Contributed to enterprise-scale static analysis frameworks for automated software verification.

PATENTS

- **Distributed Oracle Agreement System and Method (DORA)** – US Patent App. PCT/US/36923
Saurabh Joshi, Aniket Kate, Joshua Tobkin and David Yang
- **Finding Bugs with Low False Alarms and Under-specified Harness** – US Patent 8793664
Saurabh Joshi, Shuvendu Lahiri and Akash Lal

PUBLICATIONS & RESEARCH

- **“LLOR: Automated Repair of OpenMP Programs”** Utpal Bora, *Saurabh Joshi*, Gautam Muduganti and Ramakrishna Upadrasta, VMCAI 2025

- **“GPURepair: Automated Repair of GPU Kernels (Extended Version)”** *Saurabh Joshi* and Gautam Muduganti, *Sadhana*, Vol. 49, 2024.
- **“Oracle Agreement: From an Honest Super Majority to Simple Majority”** Prasanth Chakka, *Saurabh Joshi*, Aniket Kate, Joshua Tobkin and David Yang, *IEEE ICDCS* 2023.
- **“OpenMP aware MHP Analysis for Improved Static Data-Race Detection”** Utpal Bora, Shrayish Vaishay, *Saurabh Joshi* and Ramakrishna Upadrasta, *LLVM-HPC* 2021.
- **“Anomaly Detection in Data Plane Systems using Packet Execution Paths”** Archit Sanghi, Praveen Tammana, *Saurabh Joshi* and Krishna P Kadiyala, *ACM SIGCOMM SPIN* 2021.
- **“GPURepair: Automated Repair of GPU Kernels”** *Saurabh Joshi* and Gautam Muduganti, *VMCAI* 2021.
- **“On the Tractability of (k,i)-coloring (Extended Version)”** Sriram Bhyravarapu, *Saurabh Joshi*, Subrahmanyam Kalyanasundaram and Anjeneya Swami Kare, *DAM* 2020.
- **“LLOV: A Fast Static Data-race Checker for OpenMP Programs”** Utpal Bora, Santanu Das, Pankaj Kukreja, *Saurabh Joshi*, Ramakrishna Upadrasta and Sanjay Rajopadhye, *ACM TACO*, Vol. 17 (4), 2020.
- **“Phase Transition Behavior of Cardinality and XOR constraints”** Yash Pote, *Saurabh Joshi* and Kuldeep Singh Meel, *IJCAI* 2019.
- **“Open-WBO-Inc: Approximation Strategies for Incomplete Weighted MaxSAT”** *Saurabh Joshi*, Prateek Kumar, Sukrut Rao and Ruben Martins, *JSAT*, Vol. 11, 2019.
- **“Pinaka: Symbolic Execution meets Incremental Solving”** Eti Chaudhary and *Saurabh Joshi*, *TOOLympics*, *TACAS* 2019.
- **“Approximation Strategies for Incomplete MaxSAT”** *Saurabh Joshi*, Prateek Kumar, Ruben Martins and Sukrut Rao, *CP* 2018.
- **“On the tractability of (k,i)-coloring”** *Saurabh Joshi*, Subramaniam Kalyanasundaram, Anjeneya Swami Kare and Bhyravarapu Sriram, *CALDAM* 2018.
- **“Precise Predictive Analysis for Discovering Communication Deadlocks in MPI Programs”** Vojtech Forejt, *Saurabh Joshi*, Daniel Kroening, Ganesh Narayanaswamy and Subodh Sharma, *ACM TOPLAS* 2017.
- **“Equivalence Checking of a Floating-point Unit against a High-level C Model”** Rajdeep Mukherjee, *Saurabh Joshi*, Andreas Griesmayer, Daniel Kroening and Tom Melham, *FM* 2016.
- **“The Virtues of Conflict: Analyzing Modern Concurrency”** Ganesh Narayanaswamy, *Saurabh Joshi*, and Daniel Kroening, *PPoPP* 2016.
- **“On Using Incremental Encodings in Unsatisfiability-based MaxSAT Solving”** Ruben Martins, *Saurabh Joshi*, Vasco Manquinho and Inês Lynce, *JSAT*, Vol. 9, 2015.
- **“Generalized Totalizer Encoding for Pseudo-Boolean Constraints”** *Saurabh Joshi*, Ruben Martins and Vasco Manquinho, *CP* 2015.
- **“Safety Verification and Refutation by k-invariants and k-induction”** Martin Brain, *Saurabh Joshi*, Daniel Kroening and Peter Schrammel, *SAS* 2015.
- **“Property-Driven Fence Insertion using Reorder Bounded Model Checking”** *Saurabh Joshi* and Daniel Kroening, *FM* 2015.
- **“Incremental Cardinality Constraints for MaxSAT”** Ruben Martins, *Saurabh Joshi*, Vasco Manquinho and Inês Lynce, *CP* 2014.
- **“A New Method of MHP Analysis for Languages with Dynamic Barriers”** *Saurabh Joshi*, R K Shyamasundar and Sanjeev Aggarwal, *HIPS* 2012.
- **“Under-specified Harness and Interleaved Bugs”** *Saurabh Joshi*, Shuvendu Lahiri and Akash Lal, *POPL* 2012.
- **“Distributed Generalized Dynamic Barrier Synchronization”** Shivali Agrawal, *Saurabh Joshi*, and R K Shyamasundar, *ICDCN* 2011.
- **“Reactivity in SystemC Transaction-Level Models”** Frederic Doucet, R K Shyamasundar, Ingolf Krueger, *Saurabh Joshi*, and Rajesh Gupta, *HVC* 2007.

TALKS

- Invited talk on **Web3 and Program Analysis** at IICT (CompilerTech) 2024. [\[video\]](#)
- Invited talk on **Distributed Oracle Agreement with Simple Majority** at IIT Hyderabad August 2023
- Talk on **Supra Automation Network** as part of The Tech Behind Supra webinar series. [\[video\]](#)
- Invited talk on **DORA** at Ethereum Engineering Group meetup. [\[video\]](#)
- **Code Contract: What, Why and How?** at CppIndiaCon 2021. [\[video\]](#), [\[slides\]](#), [\[code\]](#)
- **GPURepair: Automated Repair of GPU Kernels**, at Qualcomm (Feb 2021).
- **Approximation Strategies for Incomplete MaxSAT**, at IIT Delhi (June 2018), CP, Lille, France (August 2018), at the fourth Indian SAT+SMT School at IIT Bombay (December 2019).

- **Generalized Totalizer Encoding for Pseudo-Boolean Constraints**, at Second Indian SAT+SMT School, Mysore Park, 2017.
- **How to establish research collaboration with researchers in your domain (India/Outside)?**, at IEEE Hyderabad section, IIIT Hyderabad, 2017.
- **Property-driven Fence Insertion using Reorder Bounded Model Checking**, at IIIT Vadodara (Sep 2015), IIT Delhi (Nov 2015), IIT Madras (May 2015), IIT Kharagpur (May 2015), IIT Hyderabad (May 2015), IIIT Delhi (May 2015), IIT Kanpur (May 2015), FM, Oslo, Norway (June 2015).
- **Little Pain Big Gain: Role of tricks in constraint solving**, at IIT Delhi (Nov 2015).
- **Interleaved Bugs and Under-specified Harness**, Mysore Park Workshop on The Future of Debugging, Mysore, 2012.
- **What is this FOSS (Free and Open Source Software) all about?**, at Krishna Institute of Technology, Kanpur, April 2010.

HONOURS AND AWARDS

- SV-COMP 2021, 2nd place in ReachSafety-Floats and ReachSafety-Loops, 3rd place in ReachSafety-Combinations (Pinaka)
- **Excellent** rating for SERB ECR grant
- SV-COMP 2020, 3rd place in ReachSafety-Floats (Pinaka)
- MaxSAT Evaluations 2019: 2 Bronze (Open-WBO-Inc)
- SV-COMP 2019, 2nd place in ReachSafety-Floats (Pinaka)
- MaxSAT Evaluations 2018: 1 Gold, 1 Silver (Open-WBO-Inc)
- MaxSAT Evaluations 2017: 2 Gold, 1 Silver (Open-WBO)
- DST Early Career Research Award 2017
- Pseudo Boolean Evaluations 2016: 2 Silver, 2 Bronze (Open-WBO)
- MaxSAT Evaluations 2015: 1 Gold, 1 Silver (Open-WBO)
- MaxSAT Evaluations 2014: 1 Gold, 1 Silver (Open-WBO)
- SVCOMP 2016: 1 Gold (2LS)

FUNDING

- **Development of Indian Telecommunication Security Assurance Requirements** – Aug 2020–Aug 2022
Co-PI, NSCS India | ₹276.79 Lakh (share: ₹60 Lakh)
- **Validating and Securing Network Protocols and NMS** – Aug 2020–Aug 2023
Co-PI, NSCS India | ₹150 Lakh (share: ₹75 Lakh)
- **Auto Grade Linux Project** – Apr 2020–Mar 2021
Consultant/Co-PI, Suzuki Motor Corporation, Japan | ₹31.97 Lakh
- **Workshop on Program Analysis and Verification** – Feb 2019
PI/Instructor, CDAC | ₹6.78 Lakh
- **Scope Enrichment of Verification Technologies** – Aug 2017–Aug 2020
Sole PI, SERB, DST, India | ₹22.5 Lakh
- **Bridging the gap between PL and verification tools** – Nov 2016–Nov 2018
Sole PI, IIT Hyderabad | ₹3 Lakh

MENTORSHIP

- **Ongoing:** Gautam Muduganti (PhD)
- Utpal Bora (PhD), IIT Hyderabad – Jan 2019–Oct 2022 (co-supervised with Dr. Ramakrishna Upadrasta) – Techniques for Static Detection of Data-races in OpenMP Programs
- Anilava Kundu (MTech RA) – co-supervised with Dr. Ramakrishna Upadrasta
- Akash Banerjee (MTech RA), IIT Hyderabad – Aug 2018–June 2021 – Proteus: Mitigate Control-Channel Attacks
- Archit Sanghi (MTech), IIT Hyderabad – Apr 2020–July 2021 (co-supervised with Dr. Praveen Tamma) – Anomaly Detection
- Eti Chaudhary (MTech RA), IIT Hyderabad – Aug 2017–July 2020 – Verification of Programs: Pinaka & GBMC
- Aditya Kumar & Sunil Sarode (MTech), IIT Hyderabad – Apr 2019–July 2020 (co-supervised with Dr. Ramakrishna Upadrasta)
- Uttaran Sinha, Arpit Aggarwal, Rasika Sapate, Prateek Kumar, Sukrut Rao, Gitanjali Mannepalli, Nidhi Dhamnani, B Shreya

TEACHING

- Total of 45 credits at IIT Hyderabad and IIT Guwahati
- Instructor: Introduction to Blockchain Programming – Graduate Elective – Mar 2026–Apr 2026
- Instructor: Constraint Solving/Programming – Graduate Elective – 4 semesters
- Instructor: Software Verification – Graduate Elective – 3 semesters
- Instructor: Software Development Fundamentals – Mar 2021–Aug 2021
- Instructor: Introduction to Programming – Nov 2020–Mar 2021
- Instructor: Principles of Programming Languages II & I – Undergraduate – multiple semesters
- Instructor: Software Engineering – Undergraduate – multiple semesters including IIT Guwahati
- Class tutor for Software Verification at University of Oxford
- TA for Advanced Compiler Optimizations and Data Streaming Algorithms at IIT Kanpur

ACADEMIC SERVICE

- PhD Evaluation Committee for Dhriti Khanna, IIIT Delhi
- Organizer: SAT+SMT Schools (8th through 3rd: 2023, 2022, 2021, 2020, 2019, 2018)
- PRSG Member, MeiTy project – Dec 2019–July 2021
- Workshop organizer at CDAC Hyderabad Feb 2020, FMUpdate 2019 IIT Hyderabad
- Associate Editor (2018–2020): Sadhana, Journal of Indian Academy of Sciences
- Reviewer/PC/AEC: CompilerTech 2026/2025/2024, AAAI 2020, CAV 2019/2018/2014, ASV 2019, JPDC, IDC 2018/2016, ICEGOV 2017, ISSTA 2016, TACAS 2015, VMCAI 2015, HVC 2014, VSTTE 2013, HiPC 2010, PPOPP 2010, Hack.IN 2009

EDUCATION

Doctor of Philosophy, Computer Science *Indian Institute of Technology (IIT) Kanpur* **Jul 2007–Feb 2013** Kanpur, India

Thesis: Program verification, automated bug-finding, and concurrency analysis.

Master of Technology (Research Assistantship), Computer Science *Indian Institute of Technology (IIT) Bombay* **Jul 2003–Jul 2006** Mumbai, India

Bachelor of Engineering, Information Technology *GCET, Sardar Patel University* **Sep 1999–Jul 2003** Gujarat, India